

Hospital Chatbot for Patient Color-Coded Clinical Pathways

AI-Driven Triage and Navigation for Resource-Constrained Settings

Project Synopsis

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1. Introduction

Context: Color-Coded Triage

Hospitals utilize standardized **color-coded triage systems** to classify patients by clinical severity. This ensures that critical resources are allocated to those who need them most.

The Triage Spectrum:

- **Red:** Emergency (Immediate resuscitation).
- **Orange:** Very Urgent (Potentially life-threatening).
- **Yellow:** Urgent (Serious but stable).
- **Green:** Non-Urgent (Minor/Routine).
- **Blue:** Dead (Certified death on arrival).

The Innovation

An **AI-powered chatbot** acts as a "Virtual Triage Nurse," available 24/7. It assesses symptoms to assign these colors and directs patients to the correct department (e.g., Mortuary for Blue, Resuscitation for Red), streamlining flow before they even reach the desk.

2. Problem Statement

The LMIC Context

In many **low- and middle-income countries**, healthcare systems face:

- **Severe Staff Shortages:** Low physician-to-patient ratios lead to bottlenecks.
- **Overcrowded Waiting Rooms:** Patients waiting hours without initial assessment.
- **Resource Misallocation:** Highly skilled staff spending time on minor cases while emergencies wait.

Critical Consequences

- 1 **Undetected Deterioration:** "Yellow" patients turning "Orange" or "Red" while waiting unnoticed in the general queue.
- 2 **System Congestion:** Non-urgent ("Green") cases clogging the Emergency Department instead of visiting Outpatient Clinics.
- 3 **Patient Confusion:** Lack of clarity on where to go leads to anxiety and patients leaving without care (LWBS).

2.Problem Statement - Continued

Core Gap: Lack of an automated, accessible system to identify a patient's **color code** and guide them immediately upon arrival.

3. Objectives

General Objective

To develop an **AI-powered hospital chatbot** that assigns standardized triage colors (Red, Orange, Yellow, Green, Blue) and navigates patients through specific clinical pathways to reduce delays.

Specific Objectives:

- **Algorithm Development:** Create logic based on validated protocols (e.g., SATS/TEWS) to calculate urgency colors accurately.
- **Pathway Mapping:** Define distinct workflows for each color:
 - *Red/Orange:* Immediate alert to ER staff.
 - *Yellow:* Direction to Urgent Care Zone.
 - *Green:* Direction to Pharmacy/OPD.
 - *Blue:* Protocol for sensitivity and transfer to Mortuary.
- **Accuracy Evaluation:** Compare chatbot color assignments against human nurse triage to assess **sensitivity** and safety.

3. Objectives - Continued

- **Impact Measurement:** Assess reduction in **Time to Provider** and overall waiting room congestion.

4. Methodology

Study Design & Setting

- **Design:** Mixed-methods approach.
 - Phase 1: Agile Development of the Chatbot.
 - Phase 2: Comparative Accuracy Assessment.
- **Setting:** District or Regional Hospital in an LMIC setting utilizing standard color-coded triage.

Data Collection & Analysis

- **Algorithm Validation:** Retrospective testing using historical patient vitals to check if the AI predicts the same color as the doctors did.
- **Usability Testing:** Surveys (SUS Scale) with patients to ensure the chatbot is easy to use for non-tech-savvy users.
- **Key Metrics:**
 - Sensitivity (catching sick patients).
 - Reduction in Waiting Times.
 - Rate of "Left Without Being Seen".

5. Significance

Implementation of the “Virtual Triage Nurse” aims to transform hospital flow:

① **Decongest the Emergency Department:**

By accurately routing **Green** patients to clinics and **Blue** cases to appropriate facilities, the ER beds remain open for true emergencies.

② **Improve Patient Safety:**

Immediate flagging of **Red/Orange** patients ensures they bypass the queue, reducing mortality associated with waiting delays.

③ **Standardize Care Delivery:**

Providing consistent, algorithmic assessment 24/7 removes human variability and fatigue from the initial sorting process.